

Process & Decision Document

1. Short Description

For this Side Quest, I worked on modifying Week 9 - Example 3 by adding a debug system to support game development and testing.

The main goal of this task was to create a debug screen that allows toggling different game features in real time. I implemented a moon gravity toggle, which switches between normal gravity and low gravity (moon-like physics). I also added an additional debug feature to display hitboxes for the player and ground sensor.

The debug system includes:

- A toggle for moon gravity (low gravity vs normal gravity)
- A toggle for showing hitboxes (player and ground sensor)
- A reset function to reposition the player
- A debug panel UI displaying current states (velocity, grounded state, attack state)

The main technical focus was:

- Modifying physics values dynamically (gravity switching)
- Handling keyboard input for debug controls
- Creating a simple on-screen debug interface
- Visualizing invisible game elements (colliders and sensors)

2. One Small Decision I Made

Originally, I considered only implementing the required moon gravity toggle. However, I decided to add an additional debug feature that displays hitboxes for the player and ground sensor.

This decision:

- Made collision behavior easier to understand visually
- Helped identify issues with grounding and movement
- Improved the usefulness of the debug system for future development

Another small decision was to include a reset key (**R**) to quickly reposition the player. This prevents the need to restart the entire game during testing, making debugging more efficient.

I also chose to display real-time information such as velocity, grounded state, and attack state in the debug panel. This helps monitor how the game responds to different physics settings, especially when moon gravity is enabled.

3. GenAI Use Statement

GenAI was used to assist with:

- Designing the debug system structure
- Implementing gravity toggling logic

All generated code was reviewed, tested, and modified to ensure understanding and correct functionality.

Appendix: GenAI Transcript

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📄 sketch.js
JavaScript

Please organize the system so that it is easy to expand with additional debug features later. I would like the controls to be keyboard-based and the debug UI to be clear and minimal.

Thought for a few seconds >

Yes — a good way to organize this is to separate the **debug system state**, **debug controls**, and **debug rendering** into one small module-like object. That makes it much easier to add new debug tools later without scattering logic across `keyPressed()`, `draw()`, and global variables. Your current file already has the right pieces, but they are spread out. 📄 sketch

↓

Here is a cleaner structure you can use in `sketch.js`:

+ |Ask anything



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